

Damián Pablo San Román Alerigi, PhD

Applied Physicist & R&D Lead — Photonics and High-Power Laser Systems

Upstream Advanced Research Center, Saudi Aramco

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Applied physicist with a decade of experience converting fundamental optical physics into deployed industrial technology. Co-leads Saudi Aramco's high-power laser programme, carried from first-principles modelling of laser–rock interaction through to the petroleum industry's **first industrial laser perforation and flowline-descaling field deployments**. Holds a patent portfolio of **26 granted inventions across eight jurisdictions** (11 pending), with sole or lead inventorship on ten, spanning laser downhole tools, hollow-core optical fibre, photonic/ML sensing, and hydrogen-sulfide-to-hydrogen valorisation.

26 / 11 PATENTS GRANTED / PENDING	8 + ~50 JOURNAL & CONFERENCE PAPERS	437 CITATIONS (GOOGLE SCHOLAR)	21 WORKS AS FIRST / LEAD AUTHOR	2 FIRST INDUSTRIAL DEPLOYMENTS
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CORE EXPERTISE

PHYSICS & ENGINEERING

- Laser–matter interaction
- Computational electromagnetics
- Nonlinear & time-varying media
- Hollow-core fibre optics
- Photonic sensing

APPLIED / DATA

- ML for subsurface inference
- Hyperspectral sensing
- Computer vision

PROGRAMME LEADERSHIP

- Lab-to-field deployment
- IP & patent strategy
- Academic–industry collaboration

TECHNICAL TOOLKIT

LANGUAGES & TOOLS

- MATLAB Python
- Fortran C
- LaTeX PETS

SOFTWARE

- COMSOL
- Mathematica RSoft

PROFESSIONAL EXPERIENCE

Applied Physicist — Co-PI, High-Power Laser Programme

2015 – Present · Upstream Advanced Research Center, Saudi Aramco, Dhahran, KSA

- Co-led the physics, modelling and field-execution programme that took high-power laser perforation and descaling from laboratory concept to the petroleum industry's **first industrial deployments** (2021–2024), documented across 25+ peer-reviewed and conference publications and a supporting patent portfolio.
- Physics & modelling foundation (2016–2019):** first-authored the governing numerical treatments of laser–rock interaction and the thermal/geomechanical dynamics of electromagnetic heating; the coupled thermal–mechanical model was published in *SPE Journal* (2019).
 - First-of-kind field deployments (2021–2024):** delivered the industry's first industrial flowline-descaling field trial (Dec. 2021) and first industrial high-power laser perforation deployment (2023), each with a companion field-validation study.
 - Diversification (2020–present):** opened three new research lines — hydrogen-sulfide valorisation to hydrogen, photonic/ML sensing for subsurface and surface assets, and hollow-core optical fibre for kilowatt-class downhole power delivery, developed with the University of Southampton Optoelectronics Research Centre.
 - Recognised by Saudi Aramco Senior Vice President Waleed A. Al-Mulhim, with the Laser Research Team, for the first laser-perforation field trial (*The Arabian Sun / Aramco Life*, 2023).
- #### Graduate Researcher, Photonics Laboratory
- 2009 – 2014 · KAUST, Thuwal, KSA · Advisor: Prof. Boon S. Ooi
- PhD research on light propagation in heterogeneous, nonlinear and time-varying media — computational electromagnetics, beam shaping and chalcogenide-glass material engineering.
 - Founding member of the KAUST Photonics Master's programme; developed a high-performance numerical toolbox for wave propagation in complex media (*Maxwell-GEMS*).

METHODS

FEM

FDTD

Spectral / paraxial methods

Inverse-problem design

EDUCATION

PhD, Electrical Engineering (Photonics)

KAUST, Saudi Arabia
2011 – 2014 · Thesis:
Exploring Heterogeneous and Time-Varying Materials for Photonic Applications, Towards Solutions for the Manipulation and Confinement of Light

M.S., Electrical Engineering (Photonics)

KAUST, Saudi Arabia —
Founding cohort
2009 – 2010

Licenciatura (B.S.), Physics

Universidad Nacional
Autónoma de México
(UNAM)
2003 – 2009 · GPA
9.67/10

IDENTIFIERS

ORCID

0000-0002-3571-6776

SCOPUS AUTHOR ID

55408065500

RESEARCHERID

N-1562-2016

LANGUAGES

Spanish (native) ·

English (fluent) · French
(working)

SELECTED INNOVATIONS (PATENTS)

US 12590836 B2 — Optical energy apparatus, systems, and methods

SOLE INVENTOR

US 12559366 B2 — Photocatalytic conversion of hydrogen sulfide to hydrogen

LEAD INVENTOR

US 12682216 B2 — Method and system for predicting properties of rock samples using thermal sensing and machine learning

SOLE INVENTOR

US 12140850 B2 + EP / WO / CN / SA — Reconfigurable optics for beam transformation

LEAD INVENTOR · 4-JURISDICTION FAMILY

US 11725504 B2 + WO / CA / EP / JP — Contactless real-time 3D mapping of surface equipment

LEAD INVENTOR · 5-JURISDICTION FAMILY

US 11555958 B2 + EP — Nested anti-resonant nodeless optical fiber and subsurface system using the same

WITH UNIVERSITY OF SOUTHAMPTON

Full record of 26 granted patents and 11 pending applications in the accompanying Extended CV.

SELECTED PUBLICATIONS

Khan *et al.*, "Photochemical valorization of hydrogen sulfide: a study of UV-induced decomposition pathways," **RSC Advances** 15(44), 37299-37308 (2025)

M. Ferree; J. Kosco; F. Laquai; A. Cavazos Sepulveda; D.P. San Roman Alerigi, "Solar Fuel Production from Hydrogen Sulfide: An Upstream Energy Perspective," **Adv. Energy Sustain. Res.** (2023)

SENIOR AUTHOR

Y. Han; Y. Fang; D.P. San Roman Alerigi; S.I. Batarseh, "Thermal/Mechanical Interaction in Laser Perforation Process: Numerical-Model Buildup and Parametric Study," **SPE Journal** 24(05), 2097-2110 (2019)

D.P. San-Roman-Alerigi; A. Ben Slimane; T.K. Ng; M. Alsunaidi; B.S. Ooi, "A possible approach on optical analogues of gravitational attractors," **Optics Express** 21(7), 8298-8310 (2013)

FIRST AUTHOR · FEATURED, LASER FOCUS WORLD

D.P. San-Roman-Alerigi; D.H. Anjum; Y. Zhang; X. Yang; A. Ben Slimane; T.K. Ng; M.N. Hedhili; M. Alsunaidi; B.S. Ooi, "Electron irradiation induced reduction of the permittivity in chalcogenide glass (As₂S₃) thin film," **J. Appl. Phys.** (2013)

FIRST AUTHOR

D.P. San-Roman-Alerigi; T.K. Ng; M. Alsunaidi; Y. Zhang; B.S. Ooi, "Generation of J0-Bessel-Gauss beam by an heterogeneous refractive index map," **JOSA A** (2012)

FIRST AUTHOR

S.I. Batarseh; D.P. San Roman Alerigi; S. Marshall; R.M. Graves, "First Industrial High-Power Laser Perforation Deployment," SPE-215016-MS (2023)

KEY MILESTONE

Full record of 12 peer-reviewed / industry journal articles and ~50 conference papers in the accompanying Extended CV.

RECOGNITION & AFFILIATIONS

- **2013** — Optics Express paper featured in Laser Focus World: Coverage of 'A possible approach on optical analogues of gravitational attractors' (Opt. Express 21, 8298).
- **2023** — Saudi Aramco Laser Research Team recognition by SVP Waleed A. Al-Mulhim: Recognition for the first industrial high-power laser perforation field trial.
- **2026** — Gulf Energy Information Excellence Awards — finalist (team/technology recognition): Two technologies named finalists: High-Power Laser Descaler Technology for Surface Flowlines (Best Pipeline Rehabilitation Technology) and WATCH: Wellsite Asset Tracking for Control of Hazards (Best Controls, Instrumentation & Automation Technology – Upstream).
- **2010** — KAUST Academic Excellence Award
- **2009** — KAUST Provost Award
- **2008** — KAUST Discovery Scholar
- **2010** — Professional memberships: IEEE, American Physical Society, Optica/OSA, SIAM, Sociedad Mexicana de Física, The Competitiveness Institute